Machine Learning-Based Phase Resolved Partial Discharge Pattern Classification using HOG and PCA

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***Abstract*—Partial Discharge (PD) detection in rotating machine insulation is crucial for preventing failures in electrical systems. This paper proposes a machine-learning method for classifying Phase-Resolved Partial Discharge (PRPD) patterns using six machine-learning classifiers, Support Vector Machine (SVM), Bagging Tree (BaggTree), K- Nearest Neighbors (KNN), Logistic Regression (LogReg), Decision Tree (DecTree), and Random Forest (RandForest). The Histogram of Oriented Gradients (HOG) is used as the feature extraction method, while Principal Component Analysis (PCA) is applied for dimensionality reduction. The models are evaluated based on their classification performance and execution time. In terms of classification accuracy, Bagging Tree achieves the highest accuracy of 89.79%, followed by Random Forest and SVM with 87.72% and 87.10%. Furthermore, KNN has the fastest execution time of 301.820 seconds, followed by Bagging Tree with 409.741 seconds. This work also investigates performance metrics such as accuracy, precision, recall, F1-score, specificity, and balanced accuracy, demonstrating the effectiveness of combining HOG and PCA with machine learning algorithms to improve PD detection. This study demonstrates the effectiveness of machine learning in enhancing PD detection, contributing to improved predictive maintenance of rotating machines.**

Keywords—PRPD classification, partial discharge, rotating machine insulation, machine learning, predictive maintenance

I. INTRODUCTION

Partial discharge (PD) is a phenomenon that occurs in high-voltage electrical equipment due to insulation defects, leading to electrical discharges in localized areas [1] [2]. Over time, these discharges can degrade the insulation material, potentially causing complete insulation failure and catastrophic damage to rotating machines such as motors and generators [3] [4]. Early detection of partial discharges is crucial for predictive maintenance, which helps avoid costly downtime and failures. One widely adopted method for monitoring PD is Phase-Resolved Partial Discharge (PRPD) pattern analysis [5], which maps discharge activity over time relative to the AC voltage waveform. The PRPD pattern provides insight into the condition of insulation, making it a valuable tool in diagnosing issues in rotating machines [6].

However, the manual interpretation of PRPD patterns is often subjective and time-consuming, relying on expert knowledge and experience [7]. Moreover, as the complexity of insulation systems increases, so does the variability in PRPD patterns, making it more challenging to accurately identify emerging fault conditions. Automated classification methods have gained significant attention to address these challenges [8]. Using machine learning (ML) techniques makes it possible to classify PRPD patterns automatically and more precisely, reducing reliance on human expertise and increasing the efficiency of PD detection systems [9].

PRPD patterns is crucial for diagnosing insulation defects in rotating machines. One of the key challenges in this process is extracting meaningful features from the complex graphical data generated by PRPD measurements [10]. To address this, the Histogram of Oriented Gradients (HOG) is known as a robust feature extraction technique [11]. HOG is particularly effective in capturing essential gradient and edge information, which highlights the structural characteristics of PRPD patterns and improves the accuracy of pattern recognition [12] [13]. However, the high dimensionality of the extracted features can increase computational complexity, potentially leading to inefficiencies in model training and testing [14]. To mitigate this, Principal Component Analysis (PCA) is employed for dimensionality reduction, ensuring that only the most relevant information is preserved. By reducing the feature space by approximately 85%, PCA enhances computational efficiency without compromising the integrity of the classification task [15] [16].

Machine learning algorithms getting popularly used in various fields, including power systems, for classification, prediction, and anomaly detection. In the context of PRPD pattern classification, Support Vector Machine (SVM) [17], Bagging Tree (BaggTree) [18], K-Nearest Neighbors (KNN) [19], Logistic Regression (LogReg) [20], Decision Tree (DecTree) [21], and Random Forest (RandForest) [22] for PRPD pattern classification. These algorithms are known for their robustness in classification tasks, each offering unique strengths. For example, SVM is effective in handling high-dimensional data, RandForest and DecTree are renowned for their capability in handling non-linear relationships, and BaggTree improves performance through model aggregation [21].

This work focuses on incorporating HOG as a feature extraction technique to enhance the pattern recognition capability, while PCA is used for dimensionality reduction, ensuring that the most important features are retained while minimizing computational complexity. The PRPD patterns extracted from rotating machine insulation are preprocessed and normalized before being fed into the ML algorithm as classification models. Each algorithm is evaluated based on its classification performance using accuracy, precision, recall, F1 score, specificity, balanced accuracy, and execution time. The systematic comparison highlights the strengths and weaknesses of each model and offers valuable insights into how machine learning can improve both the computational time and accuracy of partial discharge detection in rotating machines. The findings contribute to enhancing predictive maintenance strategies in power systems and open avenues for future advancements in automated PD detection technologies.

II. Problem Statement

One of the key challenges in PRPD classification is its reliance on expert knowledge, which limits its scalability and effectiveness in industrial applications where large-scale, real-time monitoring is needed. Fig. 1 illustrates the six defects sources originating from a rotating machine's insulation system.

As rotating machines age and operate under varying conditions, these PRPD patterns become more difficult to interpret, necessitating automated solutions that can classify them accurately and consistently. Fig. 2 shows a typical sample of the PRPD pattern, demonstrating the complexity and variability of the data that needs to be classified.

Machine learning (ML) has emerged as a powerful approach to automate PRPD pattern classification. By learning from historical data, ML models can help in identifying different types of PD based on extracted features.

Mathematically, the problem of PRPD pattern classification can be formulated as a supervised learning task. Let X=$\{x_1,x_2,\ldots,x_n\}$ represent the set of feature vectors extracted from PRPD patterns, where each x_i corresponds to a PRPD pattern with features. Let Y=$\{y_1,y_2,\ldots,y_n\}$ represent the corresponding class labels, where each y_i indicates the type of defect or discharge activity. The objective is to learn a classifier $f{:}X\rightarrow Y$ that maps feature vectors to class labels. This classifier can be constructed using various ML algorithms, and the performance can be evaluated using several performance metrics.

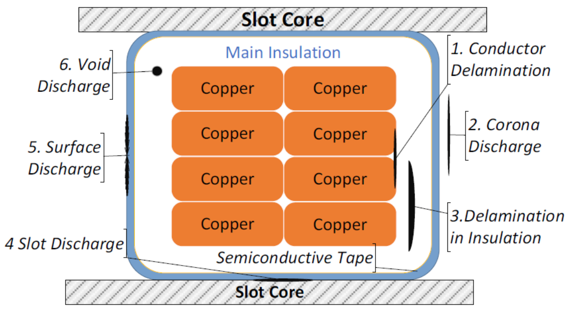

Fig. 1. Common PD Source in Rotating Machine Insulation system.

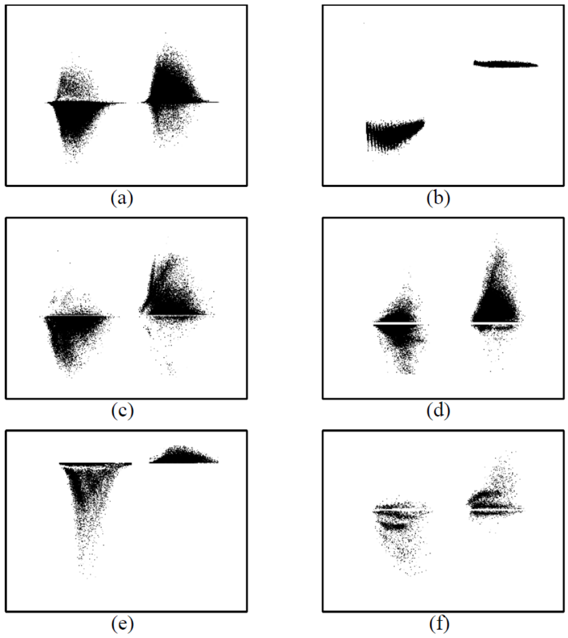

Fig. 2. Example of PRPD Pattern: (a)Conductor Delamination, (b)Corona Discharge, (c)Delamination in Insulation, (d)Slot Discharge, (e)Surface Discharge, (f)Void Discharge.

Equation (1) represents a classifier that assigns an input x to the class-y based on the highest conditional probability P(y|x). This means the model selects the class that maximizes the probability of y given x. In probabilistic classifiers, this approach ensures that the input is classified into the most likely category based on the learned data distribution. [23].

$$f(x) = argmax.P(y|x) \tag{1}$$

III. Methodology

The proposed approach integrates HOG for feature extraction, PCA for dimensionality reduction, and various ML algorithms for classification. Prior studies have shown that HOG + PCA improves classification performance over HOG alone. Here, HOG + PCA serves as a baseline to compare the effectiveness of different ML algorithms in classifying PRPD patterns.

During preprocessing, color, XY axes, and numerical labels are removed, leaving only the PRPD patterns themselves. After preprocessing, relevant features, gradient magnitudes and orientations within each cell of the image, are extracted using HOG. Next, PCA is used to reduce the feature space by 85%, optimizing computational efficiency while retaining the most relevant information for classification. Finally, several ML models are trained and evaluated to classify the PRPD patterns. The methodology flow is summarized in Fig. 3.

A. Experimental Setup and Dataset Preparation

The dataset was created using a modified simplified model of insulation defect specimens in the laboratory [24]. These specimens were tested in a high-voltage lab to replicate partial discharge conditions, with data collected under a voltage range of 1.3 to 1.5 times the Partial Discharge Inception Voltage (PDIV) threshold. This range was chosen to ensure consistency in partial discharge activity across different specimens. The experimental setup can be seen in Fig. 4.

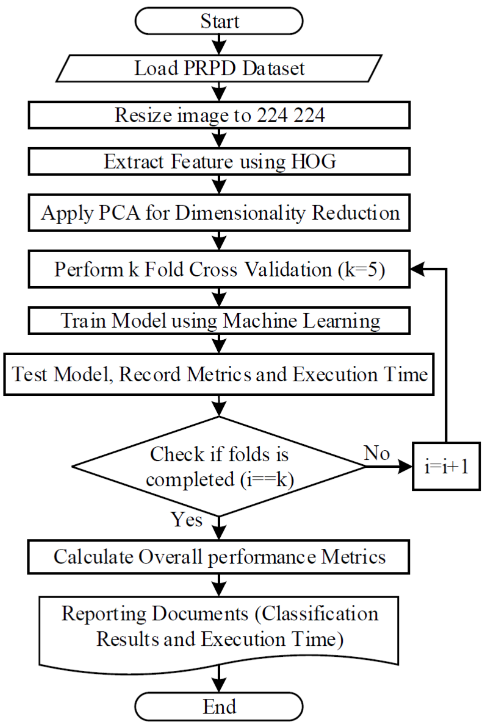

Fig. 3. Flowchart of Proposed Method.

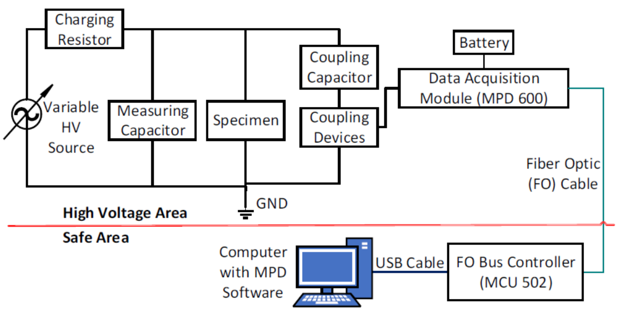

Fig. 4. Experimental Setup.

An initial dataset of 270–642 samples was generated for each defect class, capturing various patterns representative of real-world conditions. Data augmentation techniques were applied to balance the dataset across all classes and enhance model robustness, increasing the sample size to 700 PRPD images per class. This approach ensured a balanced dataset and contributed to improved model performance and generalization.

B. Histogram of Oriented Gradients (HOG) for Feature Extraction

The Histogram of Oriented Gradients (HOG) is a powerful feature extraction technique widely utilized in computer vision and image processing, particularly for object detection and recognition tasks [25]. Mathematically, the gradient at each pixel *(i,j)* in a PRPD pattern is computed as:

$$G_x(i,j) = I(i+1,j) - I(i-1,j) \tag{2}$$

$$G_y(i,j) = I(i+1,j) - I(i-1,j) \tag{3}$$

Where $G_x(i,j)$ and $G_y(i,j)$ are the horizontal and vertical gradients, and $I(i,j)$ represents the intensity of the pixel at location *(i,j)* The gradient magnitude and orientation are then calculated as:

$$Magnitude = \sqrt{G_x(i,j)^2 + G_y(i,j)^2} \tag{4}$$

$$Orientation = tan^{-1}\left(\frac{G_x(i,j)}{G_y(i,j)}\right) \tag{5}$$

Orientation histograms are created based on these gradient orientations within each cell. These histograms form the HOG descriptor, which serves as the feature vector for the subsequent machine learning models [26]

C. C. Principal Component Analysis (PCA) for Dimensionality Reduction

After extracting features using HOG, the dimensionality of the data may be high, leading to increased computational complexity and potential overfitting. PCA is employed to address this by reducing the number of features while preserving the most essential information. PCA converts the original feature set into a new principal component set that captures the data's maximum variance [27].

The PCA process begins with z-score normalizing, making the feature set X have a mean value of zero and a variance value of one. The covariance matrix Σ is then calculated as:

$$\Sigma = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^T \tag{6}$$

Where n is the total number of samples, Xi represents the feature vector of the i^{th} sample, $\bar{x}$ is the mean vector and T denotes the transpose of a matrix or vector. The eigenvectors and eigenvalues of the covariance matrix are computed, where the eigenvectors represent the principal components, and the corresponding eigenvalues indicate the amount of variance captured by each component. The principal components with the highest eigenvalues are selected, and the original data is projected onto this lower-dimensional space [28].

D. Machine Learning Algorithms for Classification

After applying PCA, the reduced feature set is fed into various ML classifiers for PRPD pattern classification. This work evaluates six ML algorithms, SVM, KNN, LogReg, DecTree, RandForest, and BaggTree. These algorithms are widely used for classification tasks. Additionally, grid search was applied to fine-tune the hyperparameters of these classifiers, ensuring optimal performance.

1. SVM constructs an optimal hyperplane in a high-dimensional space to separate classes, aiming to maximize the margin between them. It can handle non- linear relationships through kernel functions like polynomial, Gaussian, or radial basis function (RBF) kernels [29].

2. KNN is a distance-based algorithm that classifies input samples by identifying the K-nearest neighbors in the feature space and assigning the most common class among those neighbors. Euclidean distance as a distance metric determines the nearness [30].

3. Logistic Regression (LogReg) is a statistical model used to estimate the probability of a binary outcome based on one or more input features [31]. It applies the

logistic function (sigmoid curve) to map predicted probabilities to a range between 0 and 1, enabling classification into distinct categories [32].

4. Decision Tree (DecTree) is a hierarchical model that divides data into subsets based on feature values, forming a tree-like structure. Each internal node makes a decision, while the leaves represent class labels [33].

5. Random Forest (RandForest) is an ensemble learning technique that creates multiple decision trees and merges their outputs for better classification [34]. It mitigates overfitting by utilizing random subsets of both data and features, thus improving robustness and generalization [35].

6. Bagging Tree (BaggTree) is an ensemble approach where multiple decision trees are trained on randomly sampled subsets of data (with replacement). The final predictions are aggregated, often by averaging or voting, to enhance stability and reduce variance [36].

K-fold cross-validation is used to ensure the ML models generalize well to unseen data. Performance is evaluated with metrics from the confusion matrix, including accuracy, precision, recall, specificity, F1-score, and balanced accuracy. Model execution time is also tracked to assess computational efficiency. All computations are run on a system with an AMD Ryzen 5600G, 32GB RAM, and NVIDIA RTX 3060 GPU, ensuring sufficient power for model training and testing.

IV. Result and Discussion

A. Evaluation Metrics Overview

Table I summarizes the key parameters for evaluating the performance of machine learning classifiers in PRPD pattern classification. These parameters are essential for calculating performance metrics, giving a detailed assessment of each model's ability to identify partial discharge patterns.

Table II outlines the key metrics used to evaluate classifier performance, including Accuracy (correct predictions), Precision (true positives out of predicted positives), Recall (ability to detect actual positives), and F1-score (balance between precision and recall). Specificity measures true negatives out of all negatives, while Balanced Accuracy averages sensitivity and specificity, providing a more effective evaluation for imbalanced datasets.

TABLE I. Performance Metrics Parameter

Parameter	Definition
N	The overall quantity of data points
TP	Total correctly detected positive cases
TN	Total correctly detected negative cases
FP	Total incorrect positive predictions (false alarms)
FN	Total incorrect negative predictions (missed detection)

TABLE II. Metrics Definitions and Formulas for Classifier Performance Evaluation

Metric	Description	Formula
Accuracy	Proportion of correct predictions	$\frac{TP + TN}{N}$
Precision	Ratio of correct positive predictions	$\frac{TP}{TP + FP}$
Recall	Ratio of actual samples correctly predicted	$\frac{TP}{TP + FN}$
F1-Score	Balance ratio between precision and recall	$2 \text{ x } \frac{\text{Precision x Recall}}{\text{Precision + Recall}}$
Specificity	Ratio of true negatives to all negative samples	$\frac{TN}{TN + FP}$
Balanced Accuracy	Mean of sensitivity and specificity	$0.5 \text{ x}(\frac{TP}{TP + FN} + \frac{TN}{TN + FP})$

B. Classification Accuracy and Model Performance

Fig. 5 illustrates the classification performance of the ML models, evaluated using a combination of HOG and PCA. BaggTree achieved the highest accuracy at 89.79%, followed closely by RandForest at 87.72% and SVM at 87.10%. These results emphasize the effectiveness of ensemble methods like BaggTree and RandForest, which combine multiple decision trees to form stronger classifiers. Their ability to manage complex, high-dimensional data makes them ideal for PRPD pattern classification, where variability and noise in the data can significantly affect accuracy.

KNN and DecTree showed lower accuracy, with KNN reaching 74.44% and DecTree achieving 77.71%. KNN's performance may have been hindered by its sensitivity to noise and its reliance on distance-based classification, which often struggles in high-dimensional spaces. LogReg performed relatively well, with an accuracy of 86.70%, but still fell short of the ensemble methods, likely due to its assumption of linear separability, which may limit its effectiveness in the non-linear classification of PRPD patterns.

The superior performance of BaggTree and RandForest can be attributed to their ability to handle noisy and imbalanced data, as well as their capacity to model complex, non-linear relationships within the feature space. The use of multiple trees reduces overfitting, improves generalization, and ultimately results in better overall accuracy, making these models particularly suitable for industrial applications where high accuracy in detecting partial discharges is essential. A comparison between HOG alone with BaggTree and HOG PCA combined with other algorithms, such as KNN or DecTree, shows that HOG alone achieves a higher overall accuracy, reaching 78.45%. This outcome occurs because combining HOG with BaggTree proves powerful, surpassing the accuracy achieved when using HOG PCA with KNN or DecTree. However, when using the same classifier (BaggTree), HOG+PCA has been shown to improve accuracy compared to using HOG alone.

C. C. Execution Time and Computational Efficency

The computational efficiency of the machine learning models was evaluated based on their execution time for both training and testing over 5 folds, as shown in Fig. 6. The execution time represents the time taken by the model to

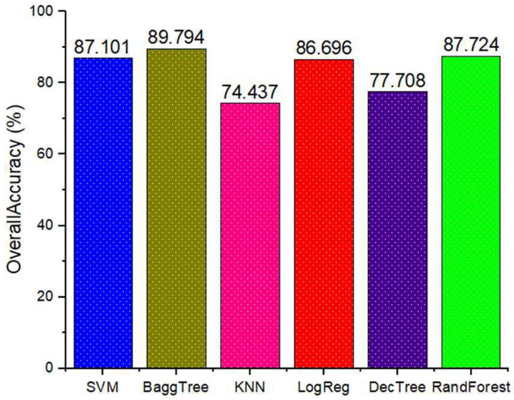

Fig. 5. Overall Accuracy Comparison.

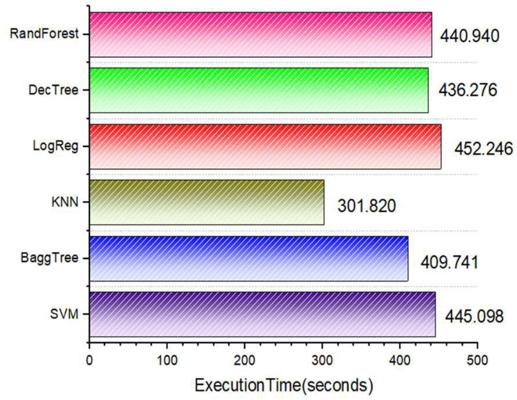

Fig. 6. Execution Time Comparison.

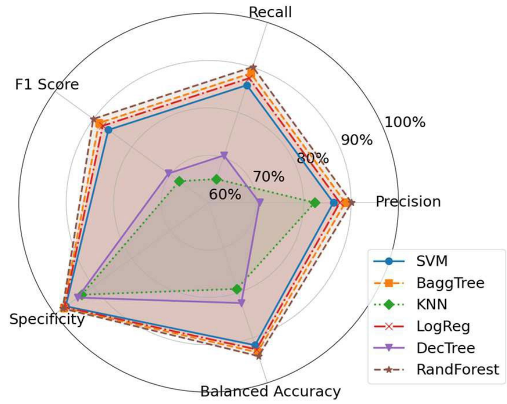

Fig. 7. Performance Metrics Radar Chart.

achieve its best accuracy, with multiple trials conducted and the highest accuracy recorded. The combination of HOG and PCA before the ML classifier played a crucial role in optimizing execution time while preserving accuracy. HOG transformed the PRPD patterns into structured feature vectors by capturing key gradient and edge information, while PCA further reduced the feature space, enabling faster processing without compromising performance.

KNN exhibited the shortest execution time at 301.82 seconds, benefiting from its simplicity as a non-training, distance-based algorithm. However, its lower accuracy limits its practicality for PRPD pattern classification. In contrast, BaggTree achieved an impressive accuracy of 89.79% with a reasonable execution time of 409.74 seconds, enabling its to use in real-time deployment. Although SVM and RandForest demonstrated competitive accuracy, their execution times were longer due to their more complex optimization and ensemble processes.

The inclusion of PCA significantly reduced the computational load by condensing the HOG-extracted features, especially for more complex models like RandForest and BaggTree, which involve multiple iterations of training. PCA ensured that the classifiers focused on the most relevant features, minimizing unnecessary computations and enhancing overall efficiency. The combination of HOG and PCA thus provided a balanced trade-off between high classification accuracy and reduced execution time, making it particularly effective for complex models in PRPD pattern classification.

D. Performance Metrics and Practical Implications

In addition to classification accuracy, five key performance metrics, precision, recall, F1-score, specificity, and balanced accuracy, were evaluated to better understand each model's capabilities, as depicted in Fig. 7. The radar chart clearly illustrates that Bagging Tree and RandForest performed the best across all metrics, with Bagging Tree achieving the highest F1-score. This suggests that these ensemble models provide a strong balance between precision and recall, ensuring reliable detection of partial discharges while minimizing both false positives and false negatives.

Bagging Tree and RandForest achieved the highest precision, minimizing false alarms crucial in industrial settings by avoiding unnecessary maintenance. Their strong recall ensures most true faults are detected, while high F1-scores show a balanced performance, effectively managing both over- and under-predictions in PRPD classification.

The high specificity and balanced accuracy of these ensemble models highlight their strength in handling imbalanced data. Specificity helps reduce false positives by accurately identifying negative cases (normal conditions), while balanced accuracy ensures equal treatment of positive and negative classes, making these models adaptable to real-world scenarios with varied class distribution. This robust performance across key metrics makes Bagging Tree and Random Forest ideal for industrial applications, where reliable detection of insulation defects is crucial for operational efficiency and minimizing downtime.

These findings are crucial for industries using rotating machines, as accurate PRPD pattern classification enables early insulation defect detection, allowing timely maintenance that reduces machine failure risk. This enhances electrical system reliability and cuts downtime and maintenance costs. Integrating models like Bagging Tree and Random Forest into monitoring systems boosts predictive maintenance efficiency, ensuring more consistent operation.

V. Conclusion

This work highlights the effectiveness of machine learning approaches for classifying PRPD patterns generated by insulation defects in rotating machines, using HOG for feature extraction and PCA for dimensionality reduction. HOG effectively captures essential gradient and edge information from the PRPD patterns, while PCA enhances computational efficiency by reducing the feature dimensionality by approximately 85%. Among the classifiers evaluated, Bagging Tree achieved the highest accuracy at 89.79%, with Random Forest just behind at 87.72%, demonstrating their suitability for real-time partial discharge detection in rotating machine insulation systems. These results indicate that integrating machine learning models with advanced feature extraction and dimensionality reduction techniques can significantly improve the reliability and scalability of predictive maintenance strategies in industrial electrical systems. For future research, deep learning methods such as CNN could improve classification accuracy by capturing more complex and detailed patterns within PRPD data, improving HOG-based feature extraction.

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References

[1] H. A. Illias, M. A. Tunio, A. H. A. Bakar, H. Mokhlis, and G. Chen, "Partial discharge phenomena within an artificial void in cable insulation geometry: Experimental validation and simulation," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 23, no. 1, pp. 451–459, 2016.

[2] D. Fahmi, H. A. Illias, H. Mokhlis, and I. M. Y. Negara, "Particle-triggered corona discharge characteristics in air insulation under DC voltage," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 30, no. 2, pp. 658–666, 2023.

[3] G. Stone, "Recent important changes in IEEE motor and generator winding insulation diagnostic testing standards," *IEEE Trans. Ind. Appl.*, vol. 41, no. 1, pp. 91–100, 2005.

[4] G. Stone, I. Culbert, E. Boulter, and H. Dhirani, *Electrical Insulation for Rotating Machines: Design, Evaluation, Aging, Testing, and Repair*, Piscataway, NJ, USA: Wiley-IEEE Press, 2004.

[5] A. Abubakar and C. Zachariades, "Phase-resolved partial discharge (PRPD) pattern recognition using image processing template matching," *Sensors*, vol. 24, no. 11, p. 3565, 2024.

[6] G. Stone, C. Maughan, D. Nelson, and R. Schultz, "Impact of slot discharges and vibration sparking on stator winding life in large generators," *IEEE Electr. Insul. Mag.*, vol. 24, no. 5, pp. 14–21, 2008.

[7] H. A. Illias, N. Y. Ting, O. Z. Yu, M. Kando, A. M. Ariffin, and M. F. M. Yousof, "Classification of partial discharges in insulation materials via support vector machine and discrete wavelet transform," in Proc. 2021 IEEE International Conference on the Properties and Applications of Dielectric Materials (ICPADM), 2021, pp. 25-28.

[8] T. Shahsavarian, Y. Pan, Z. Zhang, C. Pan, H. Naderiallaf, and J. Guo, "A review of knowledge-based defect identification via PRPD patterns in high voltage apparatus," *IEEE Access*, vol. 9, pp. 77705–77728, 2021.

[9] W. J. K. Raymond, L. T. Sing, L. W. Kin, G. K. Meng, H. A. Illias, and A. H. A. Bakar, "Feature pruning for partial discharge classification using IndFeat and ReliefF algorithm," in Proc. 2018 IEEE 2nd International Conference on Dielectrics (ICD), 2018, pp. 1-4.

[10] S. Mantach, M. Partyka, V. Pevtsov, A. Ashraf, and B. Kordi, "Unsupervised deep learning for detecting number of partial discharge sources in stator bars," *IEEE Trans. Dielectr. Electr. Insul.*, vol. X, no. Y, pp. 1–8, 2023.

[11] S. Ghaffari, P. Soleimani, K. F. Li, and D. W. Capson, "Analysis and comparison of FPGA-based histogram of oriented gradients implementations," *IEEE Access*, vol. 8, pp. 79920–79934, 2020.

[12] J. Long, L. Xie, X. Wang, J. Zhang, B. Lu, and C. Wei, "A comprehensive review of signal processing and machine learning technologies for UHF PD detection and diagnosis (II): Pattern recognition approaches," *IEEE Access*, vol. 12, pp. 29850–29890, 2024.

[13] K. Firuzi, M. Vakilian, B. T. Phung, and T. R. Blackburn, "Partial discharges pattern recognition of transformer defect model by LBP and HOG features," *IEEE Trans. Power Del.*, vol. 34, no. 2, pp. 542–550, 2019.

[14] M. Wang, W. Fu, X. He, S. Hao, and X. Wu, "A survey on large-scale machine learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 6, pp. 2574–2594, 2022.

[15] L. Li, C. Yuepeng, Y. Guang, M. Cuimin, and Z. Huajun, "Type recognition of partial discharge source based on PCA and GWO-SVM," in Proc. 2021 33rd Chinese Control and Decision Conference (CCDC), 2021, pp. 6396-6401.

[16] M. R. Hussain, S. S. Refaat, and H. Abu-Rub, "Overview and partial discharge analysis of power transformers: A literature review," *IEEE Access*, vol. 9, pp. 64587–64605, 2021.

[17] H. Al-Katheri, M. Yousof, H. Illias, and M. Talib, "Classification of fault and stray gassing in transformer oil using SVM, NB, and KNN algorithms," in Proc. 2021 IEEE International Conference on the Properties and Applications of Dielectric Materials (ICPADM), 2021, pp. 326-329.

[18] H. Herath, J. Kumara, M. Fernando, K. Bandara, and I. Serina, "Comparison of supervised machine learning techniques for PD classification in generator insulation," in Proc. 2017 IEEE International Conference on Industrial and Information Systems (ICIIS), 2017, pp. 1-6.

[19] W. Hassan, G. A. Hussain, A. Wahid, M. Safdar, H. M. Khalid, and M. K. M. Jamil, "Optimum feature selection for classification of PD signals produced by multiple insulation defects in electric motors," *Scientific Reports*, vol. 14, no. 1, 2024.

[20] H. Janani, B. Kordi, and M. J. Jozani, "Classification of simultaneous multiple partial discharge sources based on probabilistic interpretation using a two-step logistic regression algorithm," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 24, no. 1, pp. 54–65, 2017.

[21] C.-K. Chang, C.-S. Lai, and R.-N. Wu, "Decision tree rules for insulation condition assessment of pre-moulded power cable joints with artificial defects," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 29, no. 5, pp. 1636–1644, 2019.

[22] R. Yao, J. Li, M. Hui, L. Bai, and Q. Wu, "Feature selection based on random forest for partial discharges characteristic set," *IEEE Access*, vol. 8, pp. 159151–159161, 2020.

[23] A. Apicella, A. Corazza, F. Isgrò, and R. Prevete, "Toward the improvement of probabilistic classifiers using ontologies," in Proc. 2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE), 2023, pp. 664-669.

[24] Z. Wang, C. Li, P. Peng, L. Ding, Y. Jia, W. Wang, and J. Wang, "Partial discharge recognition of stator winding insulation based on artificial neural network," in Conference Record of the 2000 IEEE International Symposium on Electrical Insulation (Cat. No.00CH37075), 2000, pp. 9-12.

[25] C. G. Dias, K. L. Rodrigues, N. C. Menegasse, W. A. L. Alves, and L. C. da Silva, "Histogram of oriented gradients for rotor speed estimation in three-phase induction motors," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, May 2023.

[26] K. Firuzi, M. Vakilian, B. T. Phung, and T. R. Blackburn, "Partial discharges pattern recognition of transformer defect model by LBP & HOG features," *IEEE Trans. Power Del.*, vol. 34, no. 2, pp. 542–550, 2019.

[27] H. Ma, J. C. Chan, T. K. Saha, and C. Ekanayake, "Pattern recognition techniques and their applications for automatic classification of artificial partial discharge sources," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 20, no. 2, pp. 468–478, 2013.

[28] Y. Khan, "Partial discharge pattern analysis using PCA and back-propagation artificial neural network for the estimation of size and position of metallic particle adhering to spacer in GIS," *Electrical Engineering*, vol. 98, pp. 29–42, 2015.

[29] X. Li, X. Wang, A. Yang, and M. Rong, "Partial discharge source localisation in GIS based on image edge detection and support vector machine," *IEEE Trans. Power Del.*, vol. 34, no. 4, pp. 1795–1802, 2019.

[30] M. M. O. Harbaji, A. H. Zahed, S. A. Habboub, M. A. AlMajidi, M. J. Assaf, and A. H. El-Hag, "Design of Hilbert fractal antenna for partial discharge classification in oil-paper insulated system," *IEEE Sensors Journal*, vol. 17, no. 4, pp. 1037–1045, 2017.

[31] N. Steinauer, K. Zhang, C. Guo, and J. Zhang, "Computational modeling of gene-specific transcriptional repression, activation and chromatin interactions in leukemogenesis by LASSO-regularized logistic regression," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 18, no. 6, pp. 2109–2122, 2021.

[32] H. Janani, B. Kordi, and M. J. Jozani, "Classification of simultaneous multiple partial discharge sources based on probabilistic interpretation using a two-step logistic regression algorithm," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 24, no. 1, pp. 54–65, 2017.

[33] A. A. Soltani and S. M. Shahrtash, "Decision tree-based method for optimum decomposition level determination in wavelet transform for noise reduction of partial discharge signals," *IET sci. meas. technol.*, vol. 14, no. 1, pp. 9–16, 2020.

[34] N. A. Muhamad, I. V. Musa, Z. A. Malek, and A. S. Mahdi, "Classification of partial discharge fault sources on SF_6 insulated switchgear based on twelve by-product gases random forest pattern recognition," *IEEE Access*, vol. 8, pp. 212659–212674, 2020.

[35] S. Djaballah, L. Saidi, K. Meftah, A. Hechifa, M. Bajaj, and I. Zaitsev, "A hybrid LSTM random forest model with grey wolf optimization for enhanced detection of multiple bearing faults," *Scientific Reports*, vol. 14, no. 1, 2024.

[36] A. Hechifa, A. Lakehal, L. Saidi, A. Nanfak, R. Kelaiaia, and M. E. A. Senoussaoui, "Improvement of transformer fault diagnosis using fuzzy rule and decision tree based on dissolved gas analysis," in Proc. 2023 1st International Conference on Renewable Solutions for Ecosystems: Towards a Sustainable Energy Transition (ICRSEtoSET), 2023, pp. 1-6.